Expressing 2-particle GF in terms of $G^{(1)}$

Consider Matsubara GF

$$G(r,\tau) = -\langle T_\tau \, \hat{\rho}(r,\tau)\, \hat{\rho}(0,0)\rangle = -\mathrm{Tr}\, \frac{e^{-\beta \hat{H}_0}}{Z}\, T_\tau\, \hat{\rho}(r,\tau)\hat{\rho}(0,0)\rangle$$

$$\hat{\rho}(r,\tau) = \psi^+(r,\tau)\,\psi(r,\tau) = \frac{1}{V}\sum_{kk'} a_k^+(\tau)\, a_{k'}(\tau)\, e^{i(k'-k)r}$$

$$a_{k'}(\tau) = a_{k'}(0)\, e^{-\xi_{k'}\tau}, \quad a_k^+(\tau) = a_k^+(0)\, e^{\xi_k \tau}$$

For $\tau > 0$ $$G(r,\tau) = -\frac{1}{V^2}\sum_{kk'}\sum_{pp'} e^{i(k'-k)r}\, e^{(\xi_k - \xi_{k'})\tau} \cdot$$

$$\cdot \langle a_k^+(0)\, a_{k'}(0)\, a_p^+(0)\, a_{p'}(0)\rangle$$

The average is with density matrix $\hat{\rho} = \dfrac{e^{-\beta \hat{H}_0}}{Z}$

where $\hat{H}_0 = \sum_q \xi_q\, a_q^+ a_q$ is diagonal (this is important) quadratic Hamiltonian.

Observe that for fermions $e^{-\beta \sum_q \xi_q a_q^+ a_q} = \prod_q e^{-\beta \xi_q a_q^+ a_q} =$

$$= \prod_q \underbrace{\left(1 + (e^{-\beta\xi_q} - 1)\, a_q^+ a_q\right)}$$

$$\hookrightarrow = \begin{cases} 1 & \text{if } \hat{n}_q = a_q^+ a_q = 0 \\ e^{-\beta\xi_q} & \text{if } \hat{n}_q = 1 \end{cases}$$

We now observe that

$$\mathrm{Tr}\left\{ \prod_q \left[1 + (e^{-\beta\xi_q} - 1)\, a_q^+ a_q\right] a_k^+ a_{k'}\, a_p^+ a_{p'}\right\} =$$

$$= \sum_n \langle n | \prod_q \left[1 + (e^{-\beta\xi_q} - 1)\, a_q^+ a_q\right] a_k^+ a_{k'}\, a_p^+ a_{p'} | n\rangle$$

where $|n\rangle = |n_1, n_2, \ldots n_\ell \ldots\rangle$ describes occupation numbers of all model $\{\ell\}$ of the system

It is then clear that there are only 3 ways leading to non-zero average: (1) $k=k'$, $p=p'$ and $k\neq p$

(2) $p=k'$, $p'=k$ and $k\neq p$

(3) $k=k'=p=p'$.

Also observe that option #3 leads to

$$\frac{1}{V^2}\sum_k \langle a_k^+ a_k a_k^+ a_k\rangle \xrightarrow[V\to\infty]{} \frac{1}{V}\int(d\vec{k})\,\langle a_k^+ a_k\rangle = 0$$

i.e. one power of volume remains present so that this contribution disappears in the $V=\infty$ thermodynamic limit.

$\Rightarrow$ need to consider #1, 2 only.

$$\Rightarrow\ a_k^+ a_{k'} a_p^+ a_{p'} = a_k^+ a_k\, a_p^+ a_p\, \delta_{kk'}\delta_{pp'}$$

$$+\ a_k^+ a_p a_p^+ a_k\, \delta_{pk'}\delta_{p'k}$$

$$G(r,\tau>0) = -\frac{1}{V^2}\sum_{kp}\Big(\langle a_k^+ a_k a_p^+ a_p\rangle + \langle a_k^+ a_k a_p a_p^+\rangle \cdot e^{i(p-k)r} e^{(\xi_k-\xi_p)\tau}\Big)$$

$$= -\frac{1}{V^2}\sum_{kp}\Big(\langle a_k^+ a_k\rangle\langle a_p^+ a_p\rangle + e^{i(p-k)r} e^{\xi_k\tau}\langle a_k^+ a_k\rangle e^{-\xi_p\tau}\langle a_p a_p^+\rangle\Big)$$

$\boxed{\text{for } \vec{k}\neq\vec{p} \text{ averages factorize!}}$

$$= -\left(\frac{1}{V}\sum_k\langle a_k^+ a_k\rangle\right)\left(\frac{1}{V}\sum_p\langle a_p^+ a_p\rangle\right) +$$

$$-\frac{1}{V}\sum_k e^{-ikr}\langle a_k^+(\tau)a_k(0)\rangle\,\frac{1}{V}\sum_p e^{ipr}\langle a_p(\tau)a_p^+(0)\rangle$$

The first term is just $\bar{n}^2$, where $\bar{n} = \frac{1}{V}\sum_k\langle\hat{n}_k\rangle$ = average density.

Next observe that $\frac{1}{V}\sum_p e^{ipr} \langle a_p(\tau)\, a_p^+(0)\rangle$

$= \langle \psi(r,\tau)\, \psi^+(0,0)\rangle$;

$$\frac{1}{V}\sum_k e^{-ikr} \langle a_k^+(\tau)\, a_k(0)\rangle = \langle \psi^+(r,\tau)\, \psi(0,0)\rangle$$

Moreover, $\langle \psi(r,\tau)\, \psi^+(0,0)\rangle = \langle T_\tau\, \psi(r,\tau)\, \psi^+(0,0)\rangle$

$\underline{(\tau>0)}$ $\langle \psi^+(r,\tau)\, \psi(0,0)\rangle = -\langle T_\tau\, \psi(0,0)\, \psi^+(r,\tau)\rangle$

Indeed, $\langle T_\tau\, \psi(\tau)\, \psi^+(0)\rangle = \theta(\tau)\langle \psi(\tau)\, \psi^+(0)\rangle - \theta(-\tau)\langle \psi^+(0)\, \psi(\tau)\rangle$

$= \langle \psi(\tau)\, \psi^+(0)\rangle$ for $\tau>0$.

And $-\langle T_\tau\, \psi(0)\, \psi^+(\tau)\rangle = -\theta(\tau)\left(-\langle \psi^+(\tau)\, \psi(0)\rangle\right)$

$-\theta(-\tau)\langle \psi(0)\, \psi^+(\tau)\rangle = \langle \psi^+(\tau)\, \psi(0)\rangle$ $\underline{\text{for } \tau>0}$

We see that we need to consider slightly modified GF:

$$G^{(2)}(r,\tau) = -\langle T_\tau\, (\hat{\rho}(r,\tau)-\bar{n})(\hat{\rho}(0,0)-\bar{n})\rangle$$

for which we found

$\hat{\rho}-\bar{n}$ describes fluctuations above average

$$G^{(2)}(r,\tau>0) = -\frac{1}{V^2}\sum_{kp} e^{i(p-k)r}\langle a_k^+(\tau)\, a_k(0)\rangle\langle a_p(\tau)\, a_p^+(0)\rangle$$

$$= -\langle \psi^+(r,\tau)\, \psi(0,0)\rangle\langle \psi(r,\tau)\, \psi^+(0,0)\rangle$$

$$= (-)\left(-\langle T_\tau\, \psi(0,0)\, \psi^+(r,\tau)\rangle\right)\left(\langle T_\tau\, \psi(r,\tau)\, \psi^+(0,0)\rangle\right)$$

$$= G^{(1)}(-r,-\tau)\, G^{(1)}(r,\tau)$$

where $G^{(1)}(r,\tau) = -\langle T_\tau\, \psi(r,\tau)\, \psi^+(0,0)\rangle$ is a single-particle GF.

$$\Rightarrow G^{(1)}(-r,-\tau) = -\langle T_\tau\, \psi(0,0)\, \psi^+(r,\tau)\rangle$$

Since density is bosonic operator $\hat{\rho} = \psi^+ \psi$,

$$G^{(2)}(r, \tau<0) = + G^{(2)}(r, \tau+\beta),$$

while $G^{(1)}(r, \tau<0) = -G^{(1)}(r, \tau+\beta)$ for fermi GF $G^{(1)}$.

Hence $G^{(2)}(r,\tau) = G^{(1)}(r,\tau)\, G^{(1)}(-r,-\tau)$ for any τ.

* This is manifestation of the <u>Wick's</u> theorem: GF of 2, or more, particles reduces to products of pairs of averages (i.e. single-particle GF's) provided that H_0 is quadratic in operators. This theorem constitutes the basis for ~~[illegible]~~ perturbation theory for GFs.

* To find $G^{(2)}_R$, we first can find $G^{(2)}_M$ and express it ~~f~~ in terms of $G^{(1)}_M$. This often is the easiest way to carry out calculations.

Now $$G(r,\tau) = \frac{1}{\beta}\sum_{\omega_n}\int(d\vec{p})\, e^{ipr - i\omega_n\tau}\, G(\vec{p},\omega_n)$$

$$\Rightarrow G^{(2)}(\vec{p},\omega_n) = \int d\tau \int d\vec{r}\, e^{-i\vec{p}\vec{r} + i\omega_n\tau}\, G^{(2)}(\vec{r},\tau) =$$

$$= \int d\tau\, d\vec{r}\, e^{-ipr + i\omega_n\tau}\, G^{(1)}(\vec{r},\tau)\, G^{(1)}(-\vec{r},-\tau) =$$

$$= \int d\tau\, d\vec{r}\, e^{-i\vec{p}\vec{r} + i\omega_n\tau}\, \frac{1}{\beta}\sum_{\omega_1}\int(d\vec{p}_1)\, \frac{1}{\beta}\sum_{\omega_2}\int(d\vec{p}_2)\cdot$$

$$\cdot\, e^{i\vec{p}_1\cdot\vec{r} - i\omega_1\tau}\, e^{-i\vec{p}_2\cdot\vec{r} + i\omega_2\tau}\, G^{(1)}(\vec{p}_1,\omega_1)\, G^{(1)}(\vec{p}_2,\omega_2)$$

$$\int d\vec{r}\, e^{-i\vec{r}(\vec{p} - \vec{p}_1 + \vec{p}_2)} = \delta(\vec{p} - \vec{p}_1 + \vec{p}_2) \qquad , \vec{p}_1 = \vec{p} + \vec{p}_2$$

$$\int_0^\beta d\tau\, e^{i\tau(\omega_n - \omega_1 + \omega_2)} = \beta\, \delta_{\omega_n - \omega_1 + \omega_2, 0}$$

$$\omega_1 = \omega_n + \omega_2$$

$$\Rightarrow G^{(2)}(\vec{p},\omega_n) = \frac{1}{\beta}\sum_{\omega_2}\int(d\vec{p}_2)\, G^{(1)}(\vec{p}_2+\vec{p}, \omega_2+\omega_n)\, G^{(1)}(\vec{p}_2,\omega_2).$$

$$G^{(2)}(\vec{p},\omega_n) = \frac{1}{\beta}\sum_{\nu_\ell}\int(d\vec{q})\, G^{(1)}(\vec{q}+\vec{p}, \nu_\ell+\omega_n)\, G^{(1)}(\vec{q},\nu_\ell)$$

where $\nu_\ell = (2\ell+1)\pi T$ is Fermi Matsubara freq.

$\omega_n = 2n\pi T$ is Bose freq.

$$G^{(2)}(r,\tau) = -\langle T_\tau\, (\psi^\dagger(r,\tau)\,\psi(r,\tau)_{-n})(\psi^\dagger(0,0)\,\psi(0,0)_{-n})\rangle$$

In the absence of interactions (which is our case for now) $G^{(2)} = G^{(1)}\, G^{(1)}$ is exact. More generally, there is also a <u>vertex</u> function.

In fact, we can write everything in terms of $\mathrm{Im}\, G_R^{(1)}$

From spectral representation

$$G_R(\omega) = \int_{-\infty}^{\infty} d\omega' \, \frac{\left(1+\varepsilon\, e^{-\beta\omega'}\right) J_1(\omega')}{\omega + i\delta - \omega'}$$

$$\Rightarrow \quad \mathrm{Im}\, G_R(\omega) = -\pi \left(1+\varepsilon\, e^{-\beta\omega}\right) J_1(\omega) .$$

But for Matsubara GF we have

$$G(\omega_n) = \int_{-\infty}^{\infty} d\omega' \, \frac{\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega')}{i\omega_n - \omega'} = -\frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \, \frac{\mathrm{Im}\, G_R(\omega')}{i\omega_n - \omega'}$$

Then we can re-write expression for $G^{(2)}(\vec{p}, \omega_n)$:

$$G^{(2)}(\vec{p}, \omega_n) = \frac{1}{\beta} \sum_{\nu_\ell} \int (d\vec{q})\, G^{(1)}(\vec{q}+\vec{p}, \nu_\ell + \omega_n)\, G^{(1)}(\vec{q}, \nu_\ell)$$

$$= \frac{1}{\beta} \sum_{\nu_\ell} \int (d\vec{q}) \int_{-\infty}^{\infty} d\varepsilon_1 \, \frac{\mathrm{Im}\, G_R^{(1)}(\vec{q}+\vec{p}, \varepsilon_1)/\pi}{i\nu_\ell + i\omega_n - \varepsilon_1} \int_{-\infty}^{\infty} d\varepsilon_2 \, \frac{\mathrm{Im}\, G_R^{(1)}(\vec{q}, \varepsilon_2)/\pi}{i\nu_\ell - \varepsilon_2}$$

Do the sum:

$$S_2 = \frac{1}{\beta} \sum_{\nu_\ell} \frac{1}{i\nu_\ell + i\omega_n - \varepsilon_1} \, \frac{1}{i\nu_\ell - \varepsilon_2} =$$

$$= \frac{1}{\beta} \sum_{\nu_\ell} \left(\frac{1}{i\nu_\ell + i\omega_n - \varepsilon_1} - \frac{1}{i\nu_\ell - \varepsilon_2} \right) \frac{1}{\varepsilon_1 - \varepsilon_2 - i\omega_n}$$

Note that $\nu_\ell + \omega_n = \pi T (2\ell + 1 + 2n) = \pi T \underbrace{(2(\ell+n)+1)}_{\text{odd}}$ is Fermi frequency.

→ To evaluate $S_1 = \frac{1}{\beta} \sum_{\nu_\ell} \frac{1}{i\nu_\ell - \varepsilon_2}$

We observe that this is just $\tau \to 0$ limit of

$$G^{(1)}(\tau) = \frac{1}{\beta} \sum_{\nu_\ell} e^{-i\nu_\ell \tau} \frac{1}{i\nu_\ell - \varepsilon_2} = -\langle T_\tau \psi(\tau) \psi^+(0) \rangle$$

However, the limit depends on the sign of (τ)!

(a) $\tau \to +0 \Rightarrow G^{(1)}(\tau = +0) = -\langle \psi(0^+) \psi^+(0) \rangle$

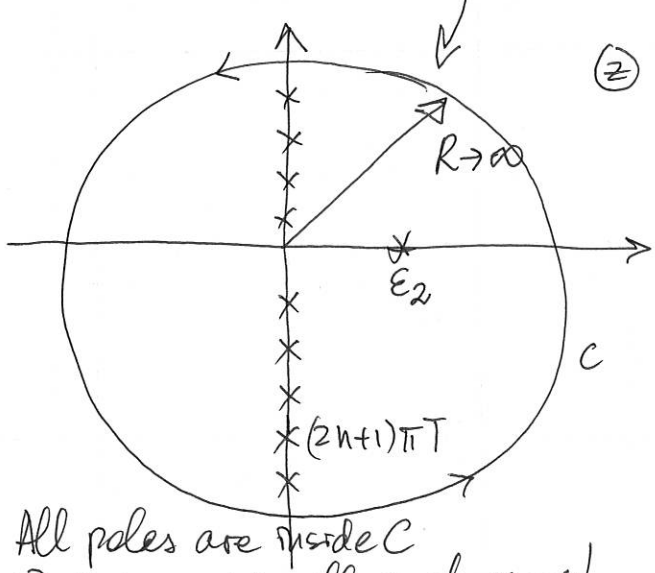
$$= (-) \{ 1 - \langle \psi^+ \psi \rangle \} = \langle \psi^+ \psi \rangle - 1.$$

(b) $\tau \to -0 \Rightarrow G^{(1)}(\tau = -0) = +\langle \psi^+ \psi \rangle.$

Thus, we regularize $S_1 \to S_1(\tau) = \frac{1}{\beta} \sum_{\nu_\ell} \frac{e^{-i\nu_\ell \tau}}{i\nu_\ell - \varepsilon_2}$

where $\lim_{\tau \to -0}$ is assumed!

Consider $\tilde{S}_1(\tau) = \oint \frac{dz}{2\pi i} \left(\frac{e^{-z\tau}}{z - \varepsilon_2} \right) \underbrace{\frac{+1}{e^{\beta z} + 1}}_{+n_F(z)}$



Poles:

$e^{\beta z_n} + 1 = 0$ for $e^{\beta z_n} = -1$

$\Rightarrow \beta z_n = (2n+1)\pi i$

$z_n = (2n+1) \frac{\pi}{\beta} i = (2n+1) \pi T i$

$$\operatorname{Res}_{z = z_n} \left(\frac{1}{e^{\beta z_n} + 1} \right) = \frac{1}{\beta e^{\beta z_n}} = -\frac{1}{\beta}$$

All poles are inside C

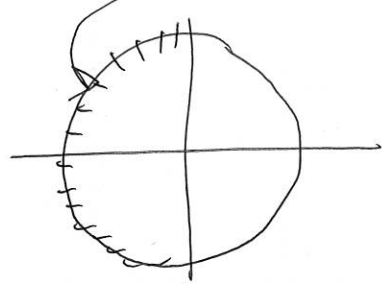
$\Rightarrow$ sum over all residues $= 0$!

$$\Rightarrow \tilde{S}_1(\tau) = \frac{2\pi i}{2\pi i} \sum_n \left(\frac{e^{-\tau z}}{z - \varepsilon_2} \right)_{z = z_n} \cdot \left(\frac{-1}{\beta} \right) + \frac{2\pi i}{2\pi i} \left(\frac{e^{-\varepsilon_2 \tau}}{e^{\beta \varepsilon_2} + 1} \right) = 0$$

Importance of the convergence factor $e^{-z\tau}$ with $\tau = -0$

$$\frac{e^{-z\tau}}{e^{\beta z}+1} \rightarrow \begin{cases} e^{-(\beta+\tau)z} & \text{for } Re(z) \rightarrow +\infty \\ e^{-z\tau} & \text{for } Re(z) \rightarrow -\infty \end{cases}$$

$\Rightarrow$ $\underline{\tau = -0}$ guarantees convergence of the contour integral on the "left" semi-circle



where $Re\{z\} = -\infty$.

Thus, $\tilde{S}_1(\tau) = \underbrace{-\frac{1}{\beta}\sum_n \frac{e^{-\tau\, i\pi T(2n+1)}}{i\pi T(2n+1)-\varepsilon_2}}_{-S_1(\tau)} + e^{-\varepsilon_2\tau} n_F(\varepsilon_2) = 0$

$\Rightarrow \quad S_1(\tau) = e^{-\tau\varepsilon_2}\, n_F(\varepsilon_2) \Rightarrow$ the limit $\tau = -0$ is easy

$\Rightarrow \quad S_1 = n_F(\varepsilon_2)$.

Hence,

$$S_2 = \frac{1}{\varepsilon_1-\varepsilon_2-i\omega_n}\left\{\frac{1}{\beta}\sum_{\ell}\frac{1}{i\nu_\ell+i\omega_n-\varepsilon_1} - \frac{1}{\beta}\sum_{\ell}\frac{1}{i\nu_\ell-\varepsilon_2}\right\}$$

(shift $\nu_\ell \to \nu_\ell - \omega_n$)

$$= \frac{1}{\varepsilon_1-\varepsilon_2-i\omega_n}\left\{\frac{1}{\beta}\sum_{\ell'}\frac{1}{i\nu_{\ell'}-\varepsilon_1} - \frac{1}{\beta}\sum_{\ell}\frac{1}{i\nu_\ell-\varepsilon_2}\right\}$$

$$= \frac{1}{\varepsilon_1-\varepsilon_2-i\omega_n}\Big(n_F(\varepsilon_1) - n_F(\varepsilon_2)\Big).$$

$$\Rightarrow \quad G^{(2)}(\vec{p},\omega_n) = \int(d\vec{q})\,\frac{1}{\pi^2}\int_{-\infty}^{\infty} d\varepsilon_1\, d\varepsilon_2\; \mathrm{Im}\, G_R^{(1)}(\vec{q}+\vec{p},\varepsilon_1)\; \mathrm{Im}\, G_R^{(1)}(\vec{q},\varepsilon_2)$$

$$\cdot\; \frac{n_F(\varepsilon_1)-n_F(\varepsilon_2)}{\varepsilon_1-\varepsilon_2-i\omega_n}$$

This can be analytically continued $i\omega_n \to \omega + i\delta$

so that $G^{(2)}(\vec{p},\omega_n) \xrightarrow[i\omega_n \to \omega+i\delta]{} G_R^{(2)}(\vec{p},\omega)$

$$G_R^{(2)}(\vec{p},\omega) = -\frac{1}{\pi^2}\int(d\vec{q})\int_{-\infty}^{\infty} d\varepsilon_1 d\varepsilon_2 \; \frac{\mathrm{Im}\, G_R^{(1)}(\bar{q}+\bar{p},\varepsilon_1)\,\mathrm{Im}\, G_R^{(1)}(\bar{q},\varepsilon_2)}{\omega+i\delta-\varepsilon_1+\varepsilon_2} \cdot \left(n_F(\varepsilon_1)-n_F(\varepsilon_2)\right)$$

$$\mathrm{Im}\, G_R^{(2)}(\vec{p},\omega) = \frac{1}{\pi}\int(d\vec{q})\int d\varepsilon_1 d\varepsilon_2 \; G_R^{(1)\prime\prime}(\bar{q}+\bar{p},\varepsilon_1)\, G_R^{(1)\prime\prime}(\bar{q},\varepsilon_2) \cdot \delta(\omega-\varepsilon_1+\varepsilon_2)\left(n_F(\varepsilon_1)-n_F(\varepsilon_2)\right)$$

$$= \frac{1}{\pi}\int(d\vec{q})\int d\varepsilon_2 \; G_R^{\prime\prime}(\bar{q}+\bar{p},\omega+\varepsilon_2)\, G_R^{\prime\prime}(\bar{q},\varepsilon_2)\left(n_F(\omega+\varepsilon_2)-n_F(\varepsilon_2)\right)$$

Finally, $G_R^{(1)}(\vec{q},\varepsilon_2) = \dfrac{1}{\varepsilon_2-\xi_q+i\delta}$

$$\mathrm{Im}\, G_R^{(1)}(\vec{q},\varepsilon_2) = -\pi\,\delta(\varepsilon_2-\xi_q)$$

$$\Rightarrow \mathrm{Im}\, G_R^{(2)}(\vec{p},\omega) = \pi\int(d\vec{q})\int(d\varepsilon_2)\;\delta(\varepsilon_2+\omega-\xi_{\bar{q}+\bar{p}}) \cdot \delta(\varepsilon_2-\xi_q)\left(n_F(\omega+\varepsilon_2)-n_F(\varepsilon_2)\right) =$$

$$= -\pi\int(d\vec{q})\;\delta(\omega+\xi_q-\xi_{q+p})\left(n_F(\xi_q)-n_F(\omega+\xi_q)\right)$$

$$= -\pi\int(d\vec{q})\;\delta(\omega+\xi_q-\xi_{q+p})\left(n_F(\xi_q)-n_F(\xi_{q+p})\right)$$

which is exactly our old result!